

Mechanochemical Synthesis of Chemically Stable Isoreticular Covalent Organic Frameworks

Bishnu P. Biswal,^{†, §} Suman Chandra,^{†, §} Sharath Kandambeth,^{†, §} Binit Lukose,[‡] Thomas Heine,[‡] and Rahul Banerjee*,[†]

[†]Physical and Materials Chemistry Division, CSIR-National Chemical Laboratory,

Dr. Homi Bhabha Road, Pune -411008.

[‡]Center for Functional Nanomaterials, School of Engineering and Science, Jacobs University Bremen, Research III, Room 61, Campus Ring 1, 28759 Bremen, Germany.

E-mail: r.banerjee@ncl.res.in; Fax: + 91-20-25902636; Tel: + 91-20-25902535.

Supporting Information

Section		Page No
S-1	General information	S2
S-2	Synthetic procedures COFs (MC) and TpBD	S3
S-3	Structural modeling and powder X-ray diffraction analysis of TpBD	S4
S-4	FT-IR spectra of TpPa-2 (MC) and TpBD (MC)	S7
S-5	¹³ C CP/MAS NMR spectra and ¹ H NMR spectra	S10
S-6	Gas and water adsorption studies	S12
S-7	Thermo Gravimetric Analysis (TGA)	S15
S-8	Scanning Electron Microscopy (SEM)	S16
S-9	HR-Transmission Electron Micrograph (HR-TEM)	S17
S-10	Stability tests of COFs (MC) and (ST)	S18
S-11	Raman spectra of all COFs (MC) and (ST)	S20
S-12	PXRD of COFs (MC) synthesized using Ball Mill (MM400) and References	S21

Section S-1: General information

General Remarks: 1, 3, 5-Triformylphloroglucinol was prepared from Phloroglucinol using literature procedure (Chong *et al.*, *Org. Lett.* **2003**, 5, 3823).¹ All other reagents and solvents were commercially available and used as received. Powder X-ray diffraction (PXRD) patterns were recorded on a Phillips PANalytical diffractometer for Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$), with a scan speed of 1° min^{-1} and a step size of 0.02° in 2θ . Fourier transform infrared (FT-IR) spectra were taken on a Bruker Optics ALPHA-E spectrometer with a universal Zn-Se ATR (attenuated total reflection) accessory in the $600\text{-}4000 \text{ cm}^{-1}$ region or using a Diamond ATR (Golden Gate). Thermogravimetric analyses (TGA) were carried out on a TG50 analyzer (Mettler-Toledo) or a SDT Q600 TG-DTA analyzer under N $_2$ atmosphere at a heating rate of $5^\circ \text{C min}^{-1}$ within a temperature range of $30\text{-}800^\circ \text{C}$. SEM images were obtained with a Zeiss DSM 950 scanning electron microscope and FEI, QUANTA 200 3D Scanning Electron Microscope with tungsten filament as electron source operated at 10 kV. The samples were sputtered with Au (nano-sized film) prior to imaging by a SCD 040 Balzers Union. HR-TEM images were recorded using FEI Tecnai G2 F30 X-TWIN TEM at an accelerating voltage of 300 kV. The TEM Samples were prepared by dropcasting the sample from isopropanol on copper grids TEM Window (TED PELLA, INC. 200 mesh). All gas adsorption experiments (up to 1 bar) were performed on a Quantachrome Quadrasorb automatic volumetric instrument and water adsorption experiments on Quantachrome Autosorb-iQ-MP automatic volumetric instrument. Solid state NMR (SSNMR) was taken in a Bruker 300 MHz NMR spectrometer and reference compound NMR data were taken in Bruker 200 MHz NMR spectrometer. All Raman measurements were carried out at room temperature on an (Jobin Yvon Horiba, France) using monochromatic radiation emitted by an Ar-laser (514.5 nm) (NRS 1500 W) operating at 20 mW using 50x long distance objective. The experiment was repeated several times and at different positions to verify the consistency of the measurement. The samples were prepared simply by putting a drop of dispersion of COF materials in isopropanol on a clean piece of Silicon wafer.

Section S-2: Synthetic procedures for COFs (MC) and TpBD

Detailed synthesis procedure:

COF-TpPa-1 (MC), TpPa-2 (MC) and TpBD (MC): 1, 3, 5-Triformylphloroglucinol (**Tp**) (0.3 mmol), aromatic diamine [p-phenylenediamine (for **TpPa-1**), 2, 5-dimethyl-p-phenylenediamine (for **TpPa-2**) and benzidine (for **TpBD**)] (0.45 mmol) was placed in a mortar (inner diameter = 3 inch or 75 mm) and with either neat or 1-2 drop of mesitylene : dioxane (1:1) grounded using pestle at room temperature, after 5 minutes a light yellow colour developed. We characterized the yellow powders via FT-IR spectra, ^{13}C solid-state NMR, and found that these yellow powders are mixture of oligomeric form with some unreacted starting materials. As time progress the colour changes to orange (15 minutes) which could be due to the increased number of units and conjugation, finally the dark red colour observed [similar to COFs (ST)] after 30 minutes age of reaction which indicates the complete COFs formation. The dark red powder collected after 45 minutes was then washed with anhydrous acetone, dichloromethane 5-6 times to remove some unreacted starting material and oligomeric impurities, then dried at 180°C under vacuum for 24 hours to give a deep red colored powder in ~90% isolated yield. FT-IR was employed to check the reaction progress, after 40 minutes of reaction all the corresponding IR peaks of COFs (MC) were matched well with the COFs (**TpPa-1**, **-2** and **TpBD**) synthesized solvothermally. **TpPa-1 (MC): IR (powder, cm^{-1});** 1609 (s), 1582 (s), 1579 (w), 1445 (s), 1256 (s), 1093 (m), 990 (s), 825 (s). **Anal. Calcd.** For $\text{C}_6\text{H}_4\text{N}_1\text{O}_1$: C, 69.92; H, 3.77; N, 13.20; found: C, 65.2; H, 3.77; N, 11.02. **TpPa-2 (MC): IR (powder, cm^{-1});** 2885 (w), 1587 (s), 1444 (s), 1250 (s), 1090 (w), 995 (s), 859 (m). **Anal. Calcd.** For $\text{C}_7\text{H}_4\text{N}_1\text{O}_1$: C, 71.18; H, 3.38; N, 11.86; found: C, 70.51; H, 5.34; N, 10.58. **TpBD (MC): IR (powder, cm^{-1});** 1594 (s), 1579 (w), 1464 (s), 1258 (s), 1093 (m), 990 (s), 825 (s). **Anal. Calcd.** For $\text{C}_9\text{H}_6\text{N}_1\text{O}_1$: C, 75.0; H, 4.16; N, 9.72; found: C, 72.51; H, 3.98; N, 8.75.

According to the suggestions of the reviewer we have performed the synthesis of **COF-TpPa-1 (MC)**, **TpPa-2 (MC)** and **TpBD (MC)** using Retsch Ball Mill (MM400). The standard experimental procedure is given below.

Synthesis of COF-TpPa-1 (MC), TpPa-2 (MC) and TpBD (MC) using Ball Mill: 1, 3, 5-Triformylphloroglucinol (**Tp**) (0.3 mmol), aromatic diamine [p-phenylenediamine (for **TpPa-1**), 2, 5-dimethyl-p-phenylenediamine (for **TpPa-2**) and benzidine (for **TpBD**)] (0.45 mmol) was

placed in a 5 mL stainless steel jar, with two 7 mm diameter stainless steel balls and either neat or 1-2 drop of mesitylene : dioxane (1:1). The mixture was ground at room temperature for 45 minutes in a Retsch MM400 mill operated at 25 Hz. The dark red powders [Red for **TpPa-1** (MC), **TpPa-2** (MC) and yellowish red for **TpBD** (MC)] collected after 45 minutes was then washed with anhydrous acetone, dichloromethane 5-6 times to remove some unreacted starting material and oligomeric impurities, then dried at 180 °C under vacuum for 24 hours to give a deep colored powder in ~90% isolated yield. Reactions were repeated to ensure reproducibility.

COF-TpPa-1 and TpPa-2: The detailed synthetic procedures for **TpPa-1** and **2** were described in our recent publication (*J. Am. Chem. Soc.* **2012**, *134*, 19524).

COF-TpBD: A Pyrex tube (o.d. $\times$ i.d. = 10 $\times$ 8 mm² and length 18 cm) is charged with 1, 3, 5-Triformylphloroglucinol (**Tp**) (63 mg, 0.3 mmol), Benzidine (**BD**) (82.9 mg, 0.45 mmol), 1.5 mL of mesitylene, 1.5 mL of dioxane, 0.5 mL of 3 M aqueous acetic acid. This mixture was sonicated for 10 minutes in order to get a homogenous dispersion. The tube was then flash frozen at 77 K (liquid N₂ bath) and degassed by three freeze-pump-thaw cycles. The tube was sealed off and then heated at 120 °C for 3 days. A red colored precipitate formed was collected by centrifugation or filtration and washed with anhydrous acetone. The powder collected was then solvent exchanged with anhydrous acetone 5-6 times and then dried at 180 °C under vacuum for 24 hours to give a deep red colored powder in 80% isolated yield. **IR (powder, cm⁻¹):** 1594 (s), 1579 (w), 1464 (s), 1258 (s), 1093 (m), 990 (s), 825 (s). **Anal. Calcd.** For C₉H₆N₁O₁: C, 75.0; H, 4.16; N, 9.72; found: C, 71.98; H, 3.33; N, 8.82.

Section S-3: Structure modeling and atomic coordinates of **TpBD**

Atomic positions and cell sizes of modeled COF layers were optimized using Self-Consistent Charge Density Functional Tight-Binding (SCC-DFTB) Method. Stacking of layers are affected by the Coulomb repulsion between the partial atomic charges in adjacent layers.² Hence, we performed Mulliken population analysis for the charges. The adjacent layers were shifted with respect to each other in different directions in order to avoid Coulomb repulsion from charges alike. Several possibilities were considered, however, the best was taken from comparison of simulated PXRD pattern with the experimental. Interlayer separation was also determined from the comparison of PXRD patterns. The fractional coordinates of **TpBD** are given in Table S1.

In order to elucidate the structure of these COFs and to calculate the unit cell parameters, possible 2-D models were optimized using Density Functional Tight-Binding method. Several stacking possibilities were considered for reasons reported in the literature.² The experimental PXRD patterns are agreeable with the simulated patterns of some near-eclipsed stacking models (Figure S1 and S2 in ESI). Hence we propose structures close to hexagonal space group (*P6/m*) for **TpBD** by comparing the experimental and simulated PXRD patterns. Refinements of PXRD pattern were done using Reflex module of Material studio.³

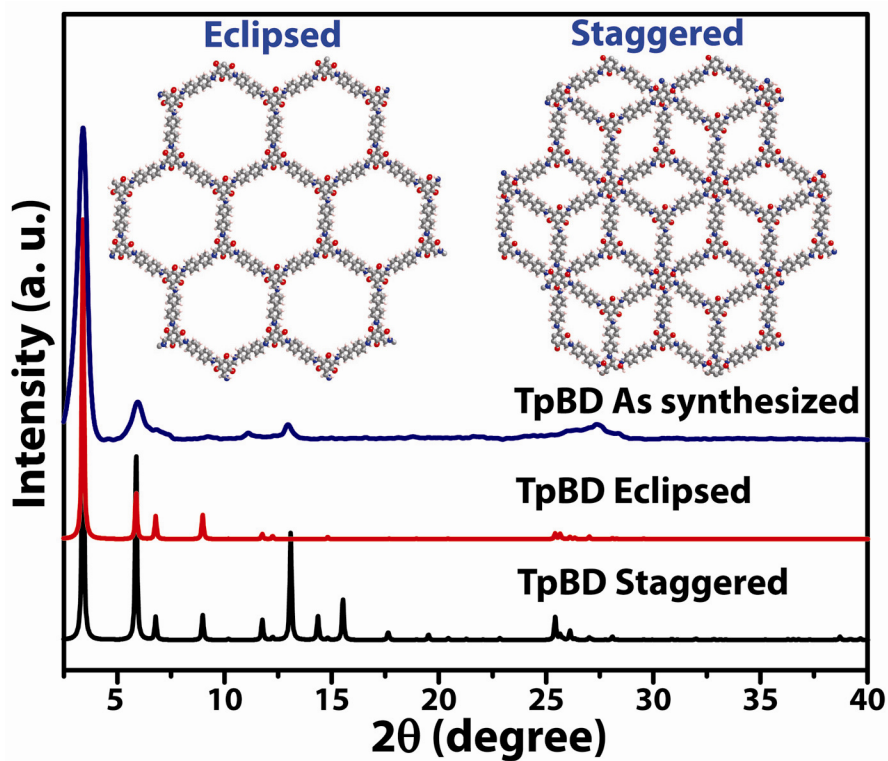


Figure S1: PXRD patterns of as-synthesized **TpBD** (Blue) compared with the Eclipsed (red) and staggered (black) stacking models.

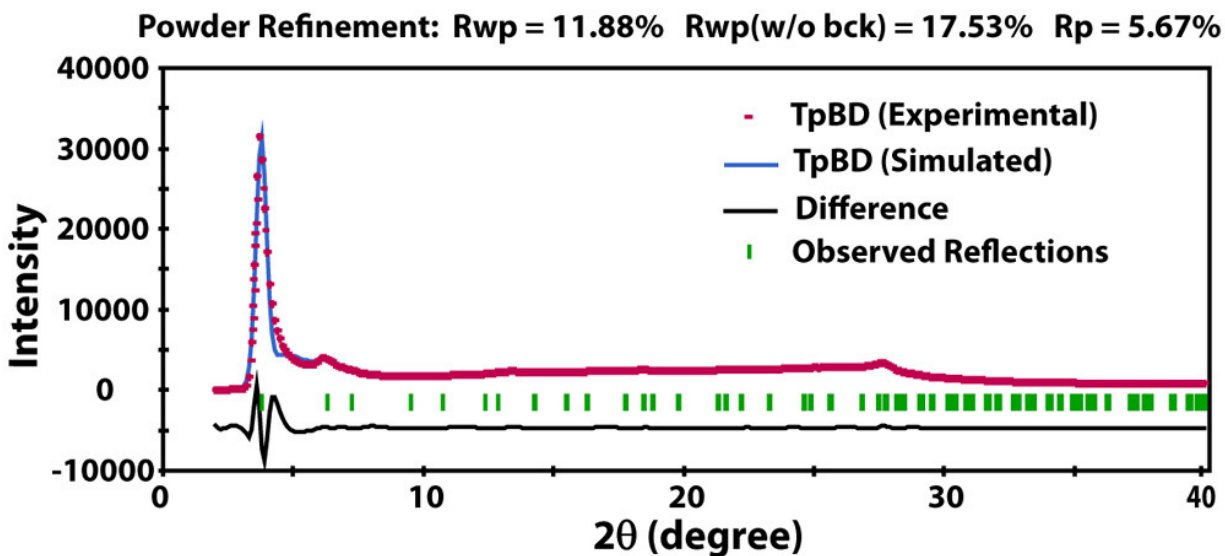


Figure S2: Experimental (Red) compared with refined (Blue) PXRD profiles of COF-TpBD with an eclipsed arrangement; difference plot is given in (black).

Space group for TpBD: <i>P6/m</i>
$a = b = 29.28710 \text{ \AA}, c = 3.25000 \text{ \AA}$
$\alpha = 90^\circ, \beta = 90^\circ, \gamma = 120^\circ$

Table S1. Fractional atomic coordinates for the unit cell of TpBD

COF-TpBD			
Hexagonal: <i>P6/m</i> $a = b = 29.28710 \text{ \AA}, c = 3.25000 \text{ \AA}$			
O	0.2956	0.5612	0
N	0.3872	0.5781	0
C	0.3135	0.6107	0
C	0.4039	0.6298	0
C	0.4195	0.5569	0
C	0.4739	0.5883	0
C	0.5046	0.5656	0
C	0.4829	0.5112	0
C	0.4281	0.4803	0
C	0.3968	0.5027	0
C	0.3684	0.6472	0
H	0.4458	0.6592	0
H	0.4926	0.6314	0
H	0.3442	0.5557	0
H	0.3534	0.4783	0
H	0.5476	0.5916	0
H	0.4091	0.4372	0

Section S-4: FT-IR Spectra

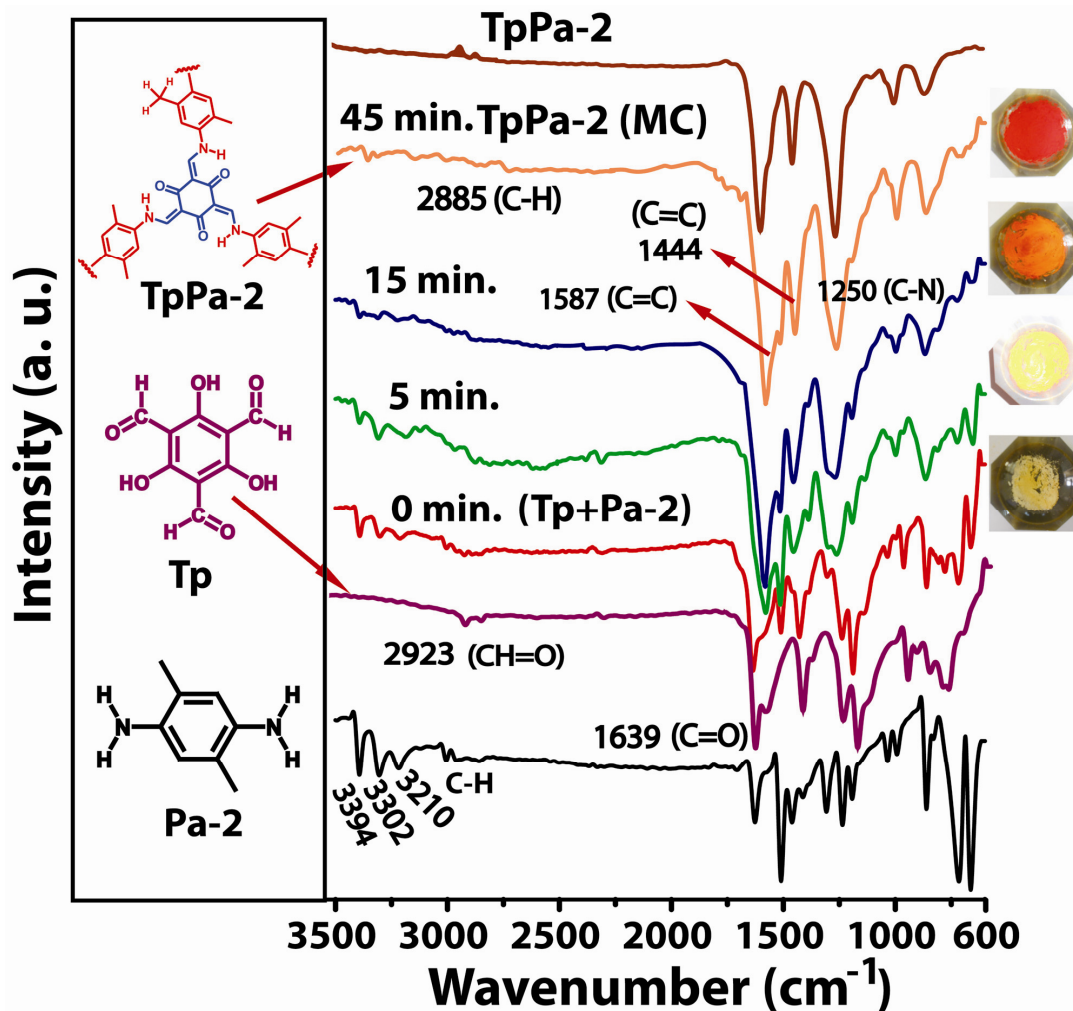


Figure S3: Stepwise comparison of the FT-IR spectra showing progress of reaction with time for **TpPa-2** (MC); black, brown, red represents [2,5-dimethyl-p-phenylenediamine (**Pa-2**)], [1, 3, 5-triformylphloroglucinol (**Tp**)], [physical mixture of **Tp** and **Pa-2**] and green, blue, golden yellow for 5, 15 and 45 minutes grounded materials respectively. Top one represents **TpPa-2** synthesized by solvothermal method (ST) (right inset images shows the change in colour observed during grinding).

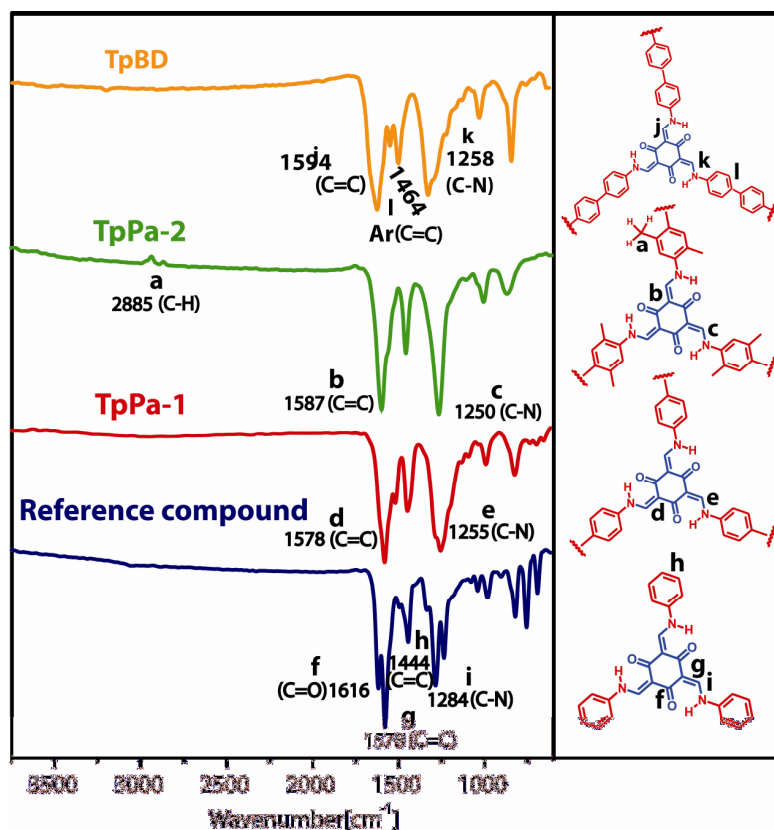


Figure S4: FT-IR spectra of **TpPa-1** (red), **TpPa-2** (green), and **TpBD** (golden yellow) with respect to the reference compound 2, 4, 6-tris[(phenylamino)methylene]cyclohexane-1, 3, 5-trione (blue).

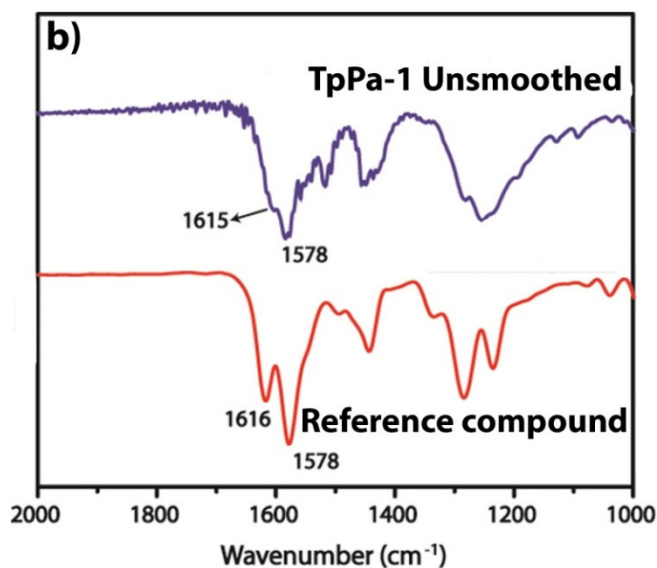


Figure S5: Unsmoothed FT-IR spectra of **TpPa-1** (blue), compared with the reference compound 2, 4, 6-tris[(phenylamino)methylene]cyclohexane-1, 3, 5-trione (red).

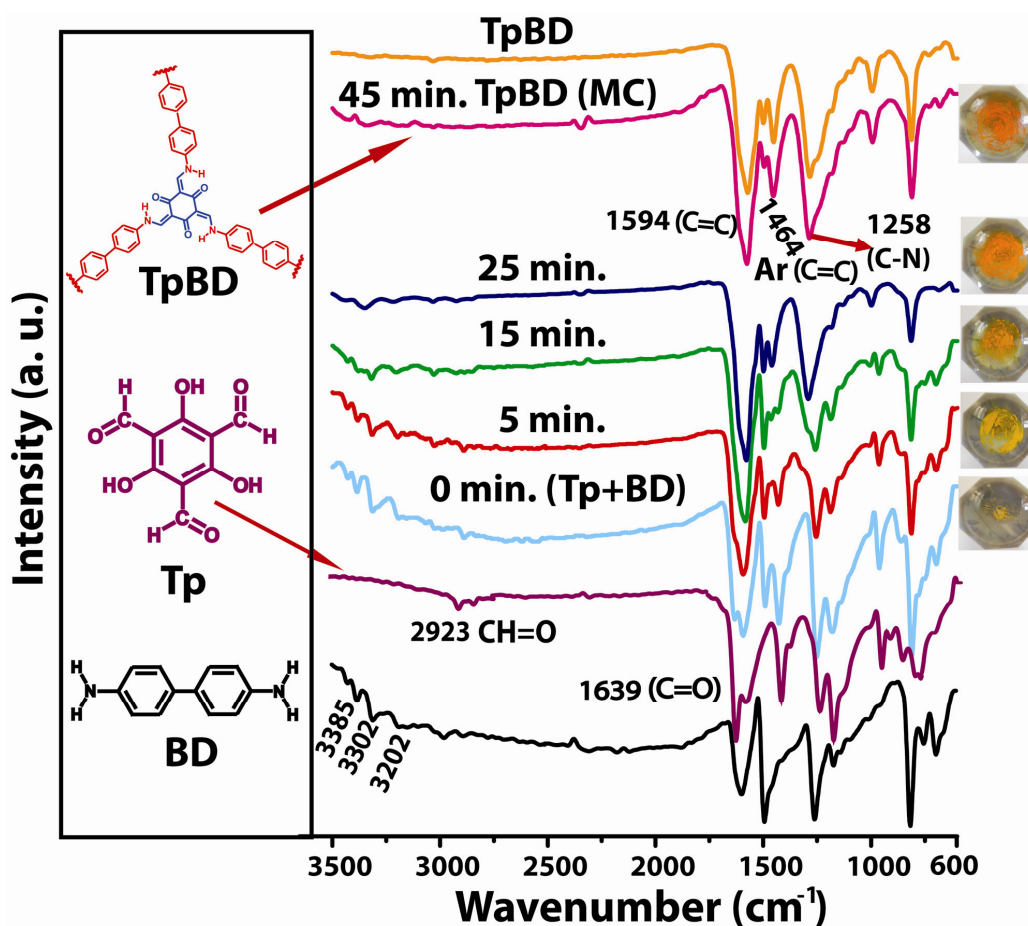


Figure S6: Stepwise comparison of the FT-IR spectra showing progress of reaction with time for **TpBD** (MC); black, brown, cyan represents [Benzidine (**BD**)], [1, 3, 5-triformylphloroglucinol (**Tp**)], [physical mixture of Tp and BD]; red, green, blue and pink for 5, 15, 25 and 45 minutes grounded materials respectively. Golden yellow (top) represents **TpBD** synthesized by solvothermal method (ST) (right inset images shows the change in colour observed during grinding).

Section S-5: ^{13}C CP/MAS NMR spectra

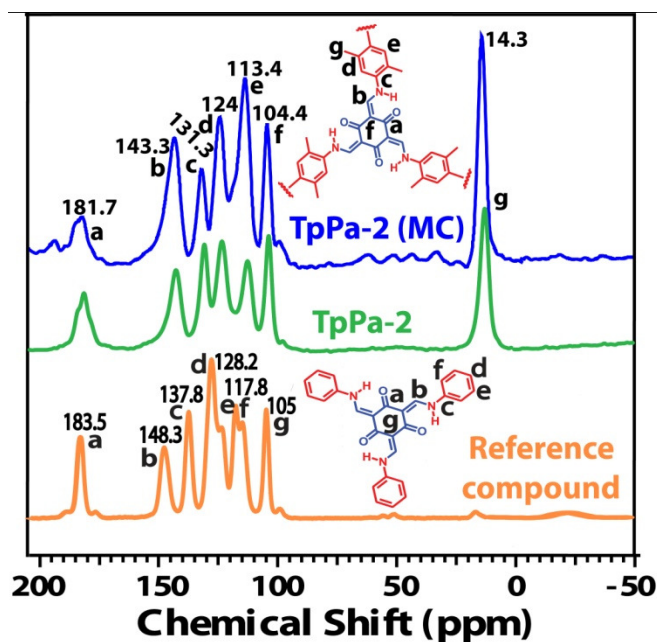


Figure S7: ^{13}C CP-MAS spectrum of **TpPa-2 (MC)** (Blue), **TpPa-2** (green) with respect to the reference material 2, 4, 6-tris[(phenylamino)methylene]cyclohexane-1, 3, 5-trione (Golden yellow).

^1H nuclear magnetic resonance spectroscopy (^1H NMR)

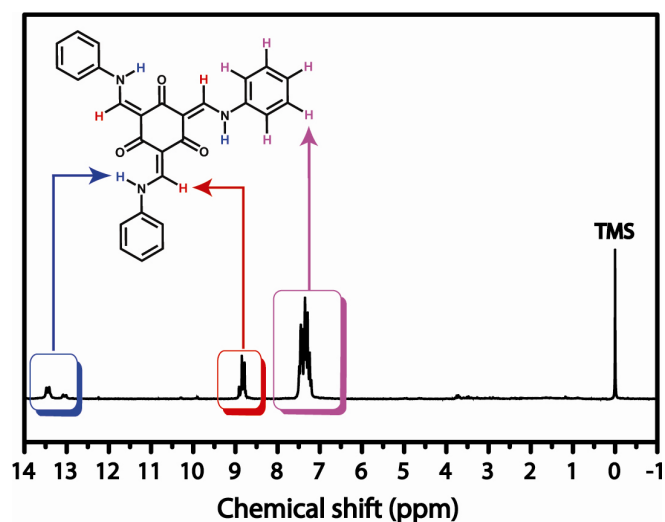


Figure S8: ^1H NMR of 2, 4, 6-tris[(phenylamino)methylene]cyclohexane-1, 3, 5-trione.

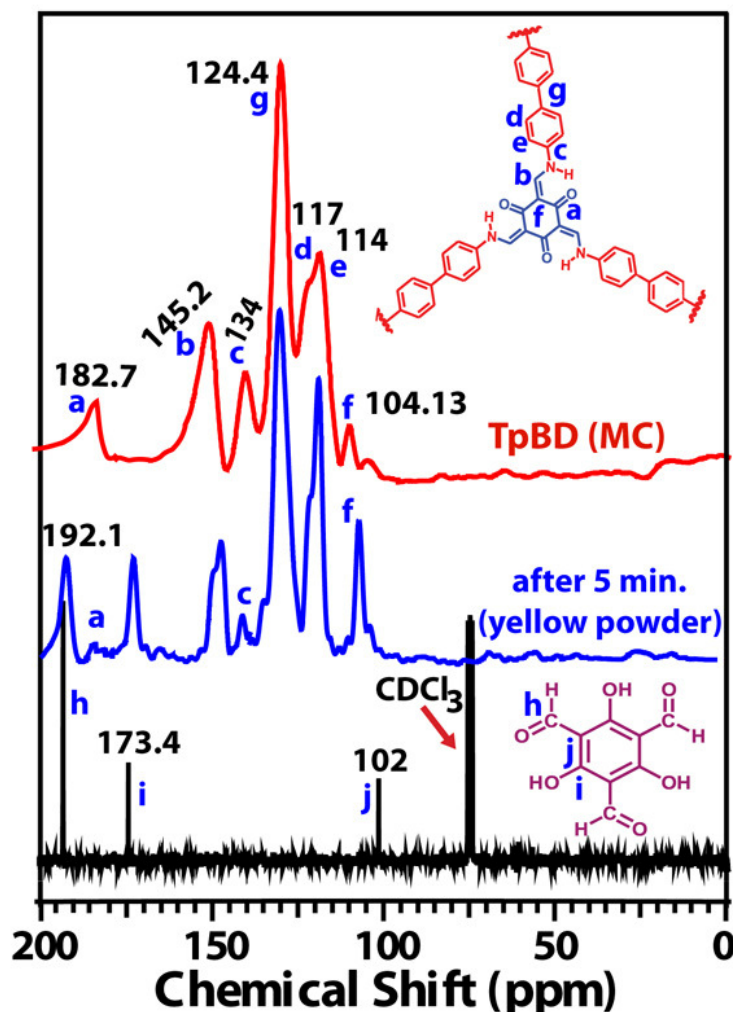


Figure S9: ^{13}C CP-MAS spectrum of **TpBD (MC)** (red), oligomers (yellow powder resulted after 5 minutes of grinding) (blue) compared with respect to the solution based ^{13}C NMR of 1, 3, 5-Triformylphloroglucinol (**Tp**) (black) taken in CDCl_3 .

Here, it has been observed that the reactions started very early (before 5 minute of grinding) but not completed, still starting materials (**Tp**) and **BD** remained along with some oligomeric form which indicated by some ^{13}C NMR peaks matched with the **TpBD (MC)** NMR spectrum. We speculate that in these small units some enol form also exists with free carbonyl group (bonded to only one or two carbonyls) at the edges which give the peaks at ~ 173.4 and 192.1 ppm.

Section S-6: Gas adsorption studies of TpPa-1, TpPa-2 and TpBD

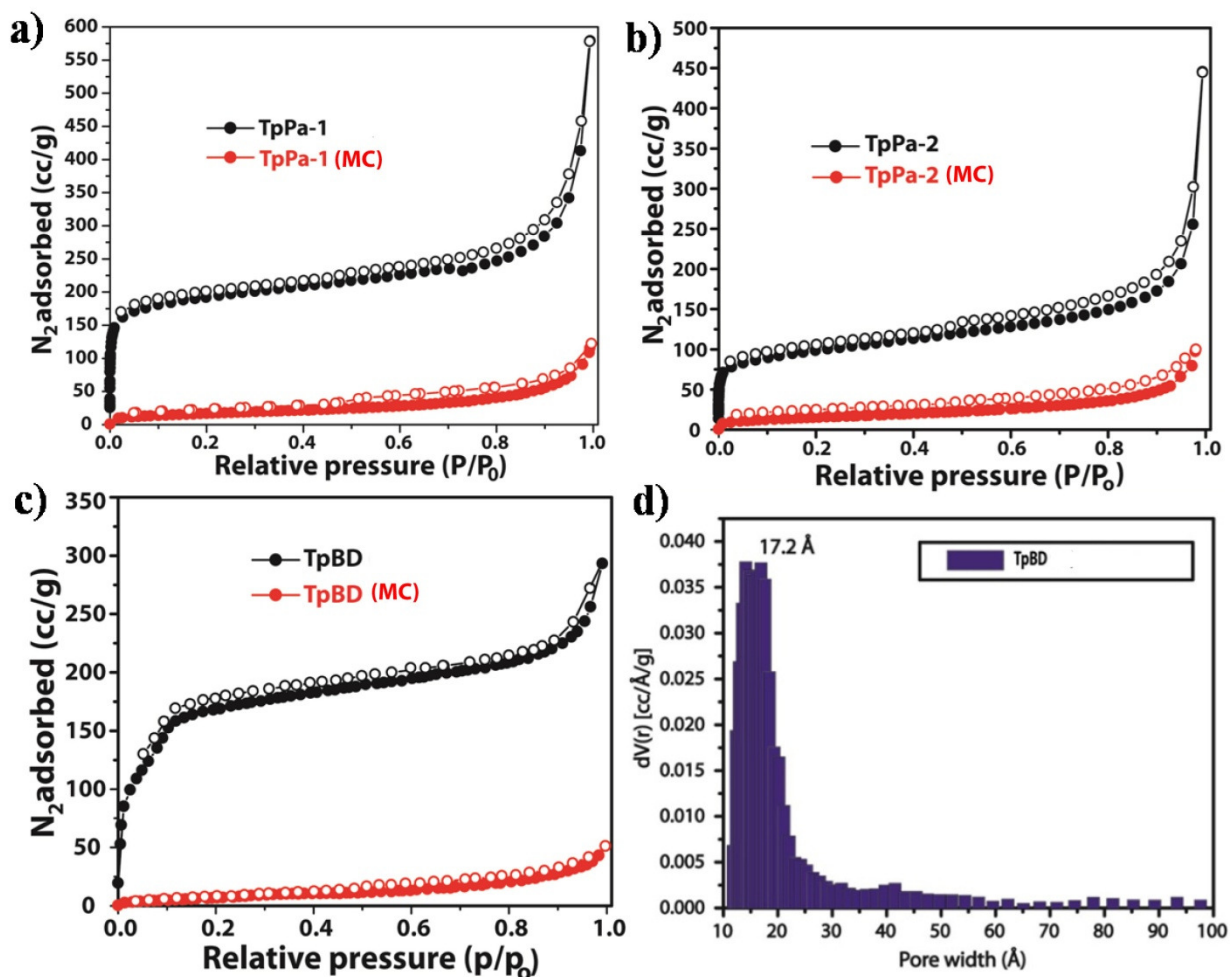


Figure S10: N_2 adsorption isotherms collected at 77 K. a) **TpPa-1** and (MC) b) **TpPa-2** and (MC) c) **TpBD** and **TpBD** (MC) and d) pore size distribution of **TpBD**. Black for COFs synthesized solvothermally (ST), red for COFs synthesized mechanochemically (MC); adsorption represented by filled circles and desorption by empty circles.

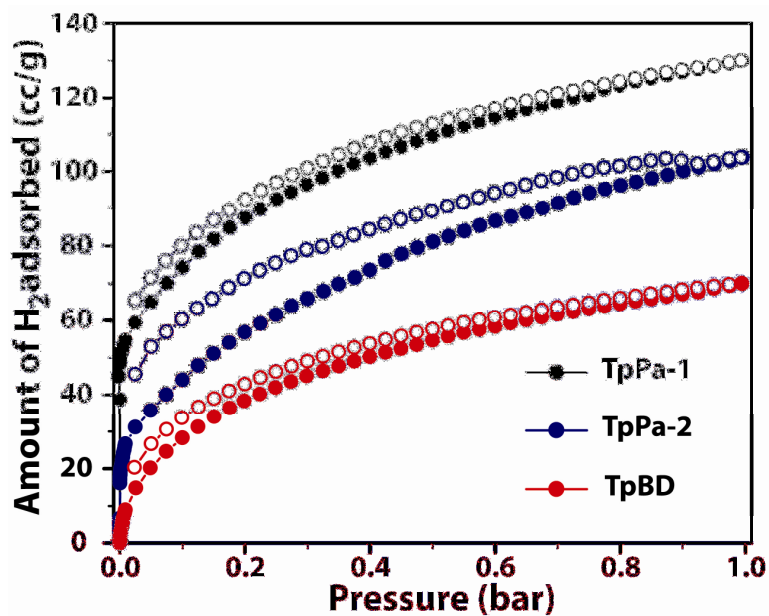


Figure S11: Hydrogen adsorption isotherms of **TpPa-1**, **TpPa-2** and **TpBD** collected at 77 K.

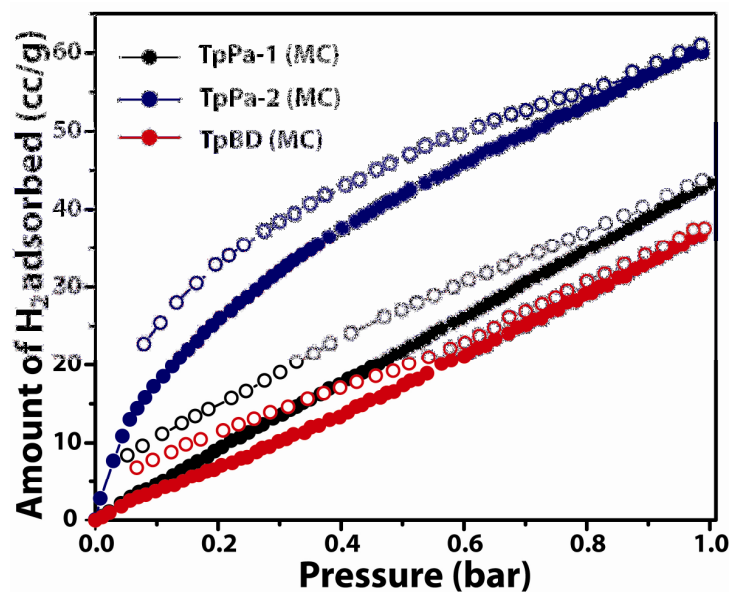


Figure S12: Hydrogen adsorption isotherms of **TpPa-1 (MC)**, **TpPa-2 (MC)** and **TpBD (MC)** collected at 77 K.

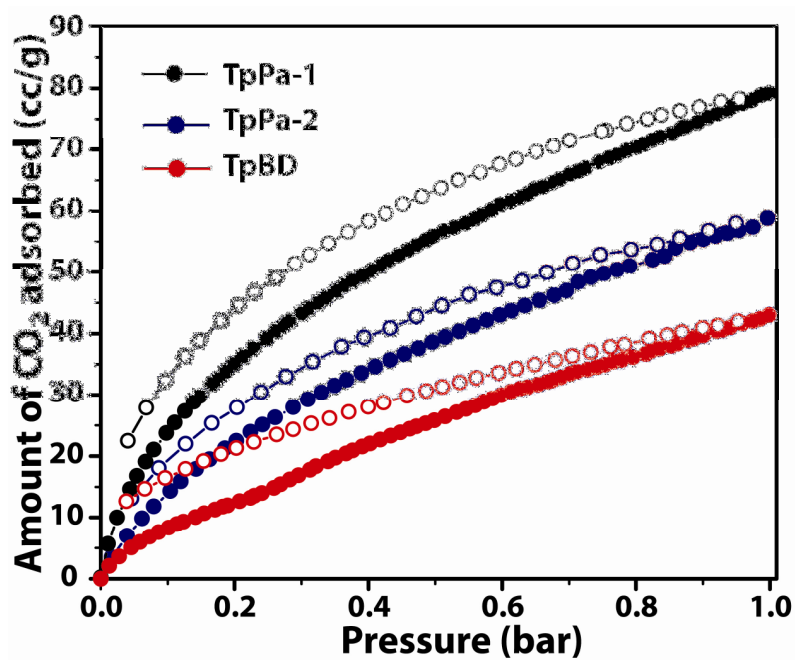


Figure S13: Carbon dioxide adsorption isotherms of **TpPa-1**, **TpPa-2** and **TpBD** collected at 273 K.

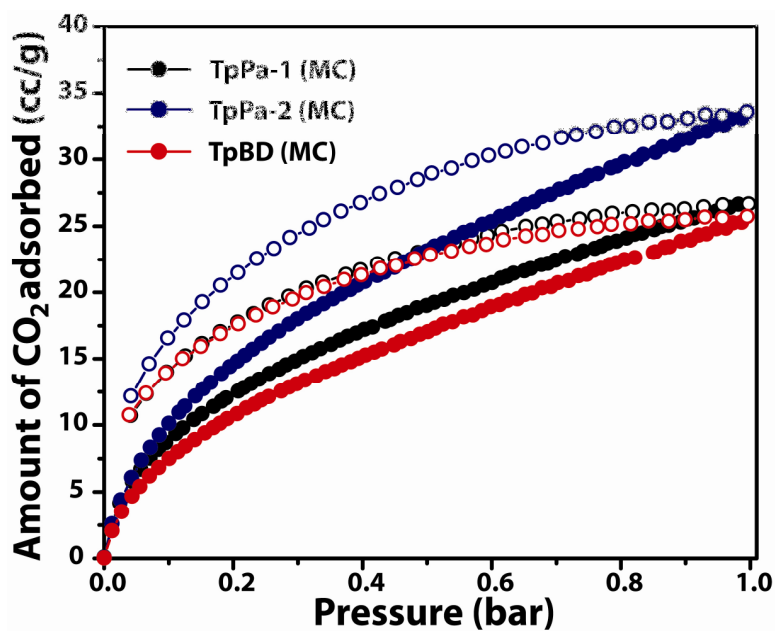


Figure S14: Carbon dioxide adsorption isotherms of **TpPa-1 (MC)**, **TpPa-2 (MC)** and **TpBD (MC)** collected at 273 K.

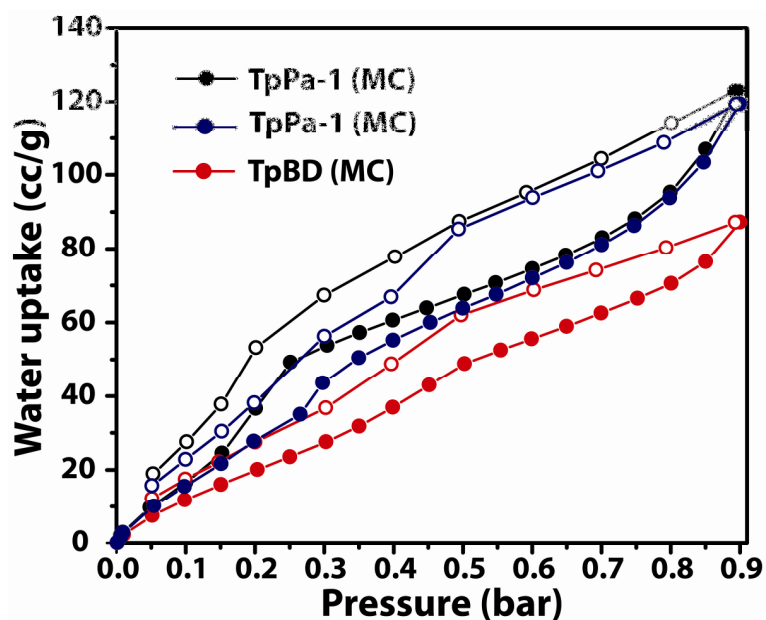


Figure S15: Water vapour adsorption isotherms collected up to $P/P_0=0.9$ at 293 K. black; **TpPa-1** (MC), blue; **TpPa-2** (MC) and red; **TpBD** (MC) synthesized via mechanochemical grinding.

Section S-7: Thermo Gravimetric Analysis (TGA)

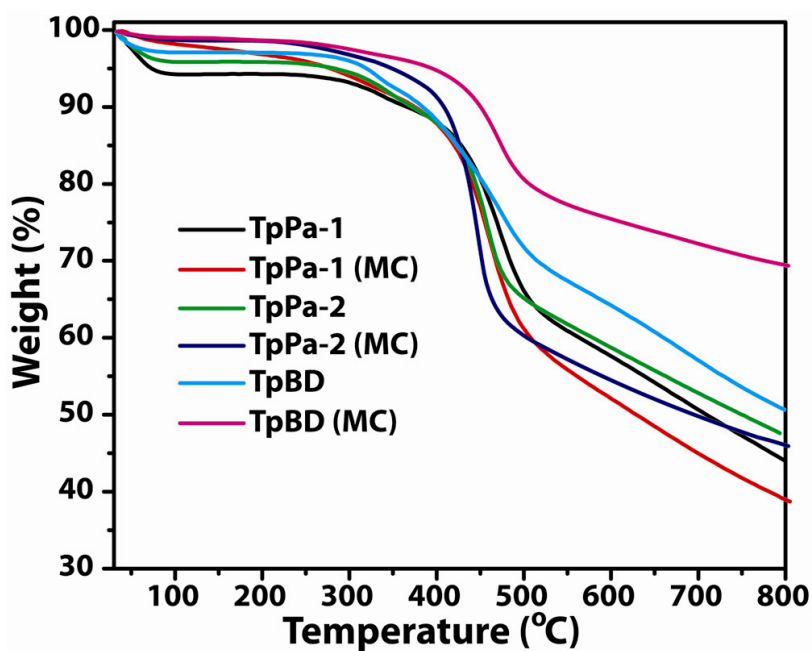


Figure S16: TGA data of all activated COFs (MC) and COFs (ST) under N_2 atmosphere.

Section S-8: Scanning Electron Micrographs (SEM)

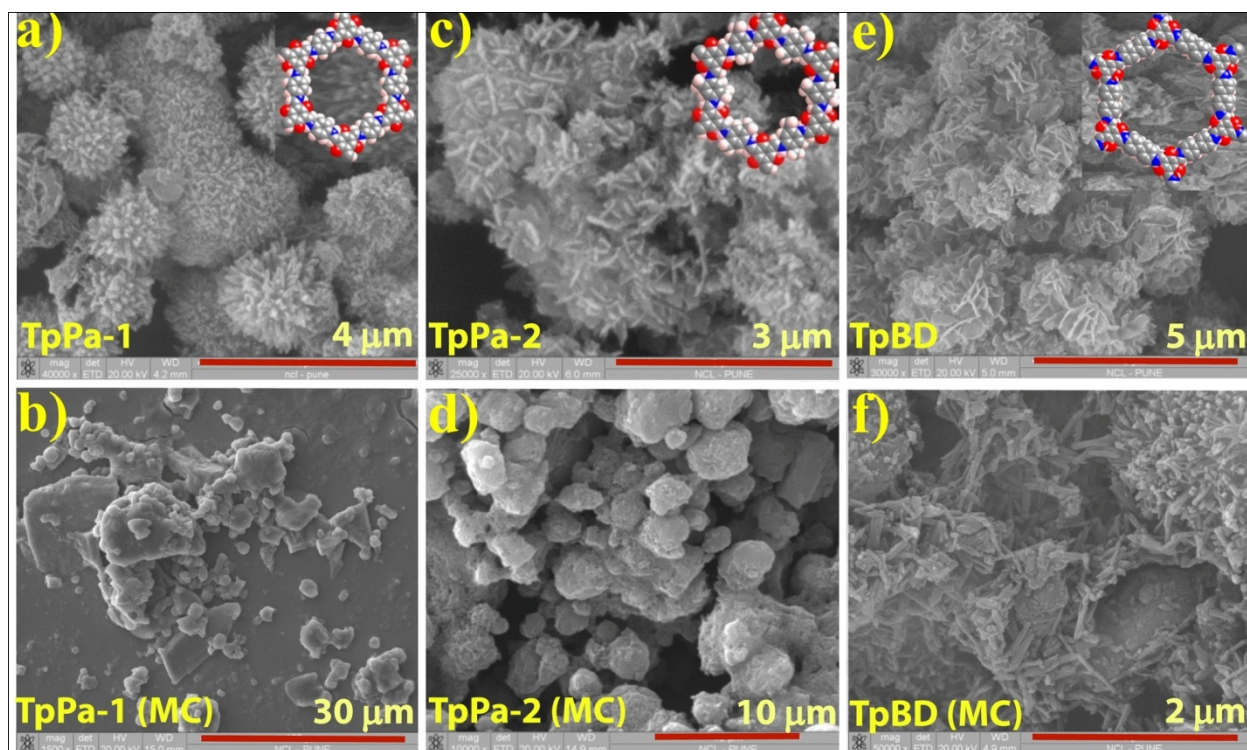


Figure S17: SEM images a), c), and e) represents **TpPa-1**, **TpPa-2**, and **TpBD** synthesized solvothermally (ST); and b), d), and f) for **TpPa-1 (MC)**, **TpPa-2 (MC)** and **TpBD (MC)** synthesized mechanochemically (MC) respectively.

Section S-9: Transmission Electron Micrographs (TEM)

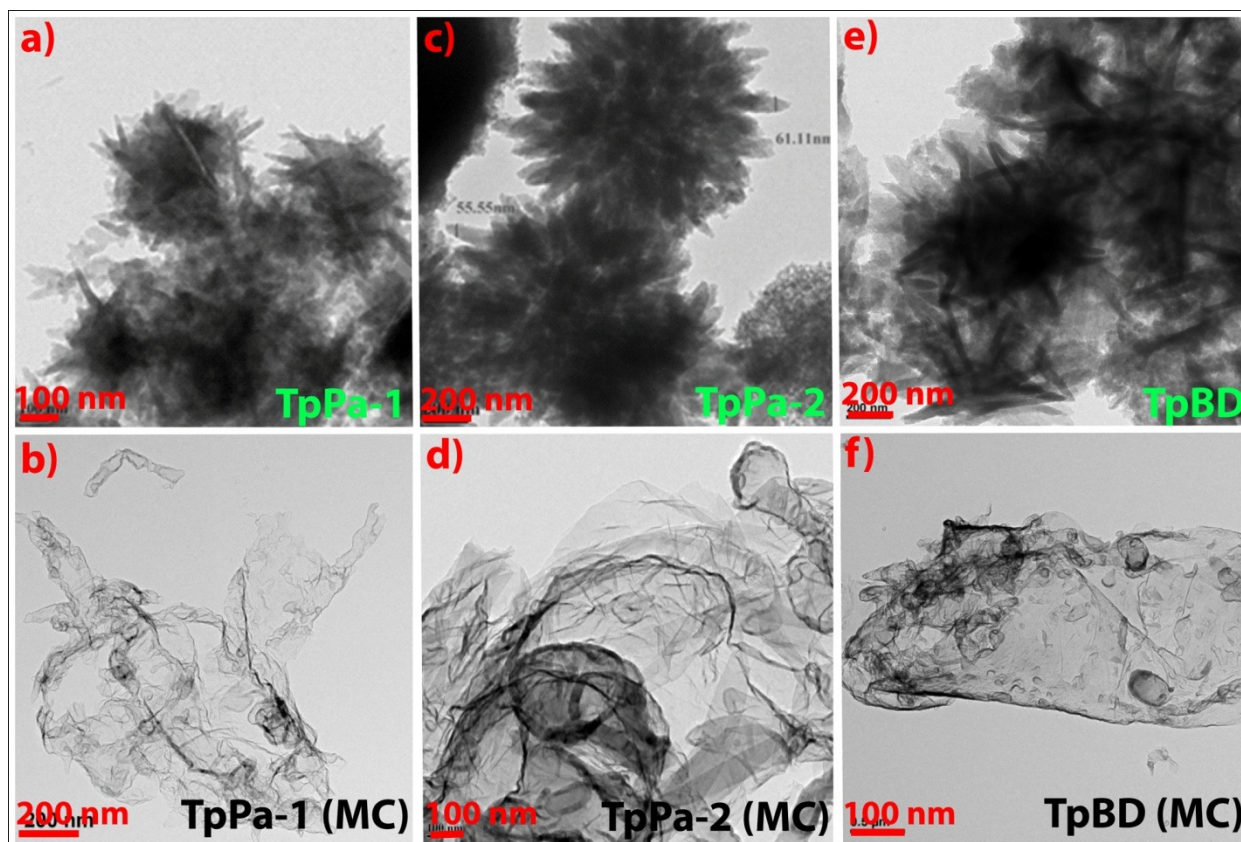


Figure S18: HR-TEM images a), c) and e) represents **TpPa-1**, **TpPa-2** and **TpBD** synthesized solvothermally (ST); and b), d) and f) for **TpPa-1** (MC), **TpPa-2** (MC) and **TpBD** (MC) synthesized mechanochemically (MC) respectively.

Section S-10: Stability tests of COFs

Stability in boiling water

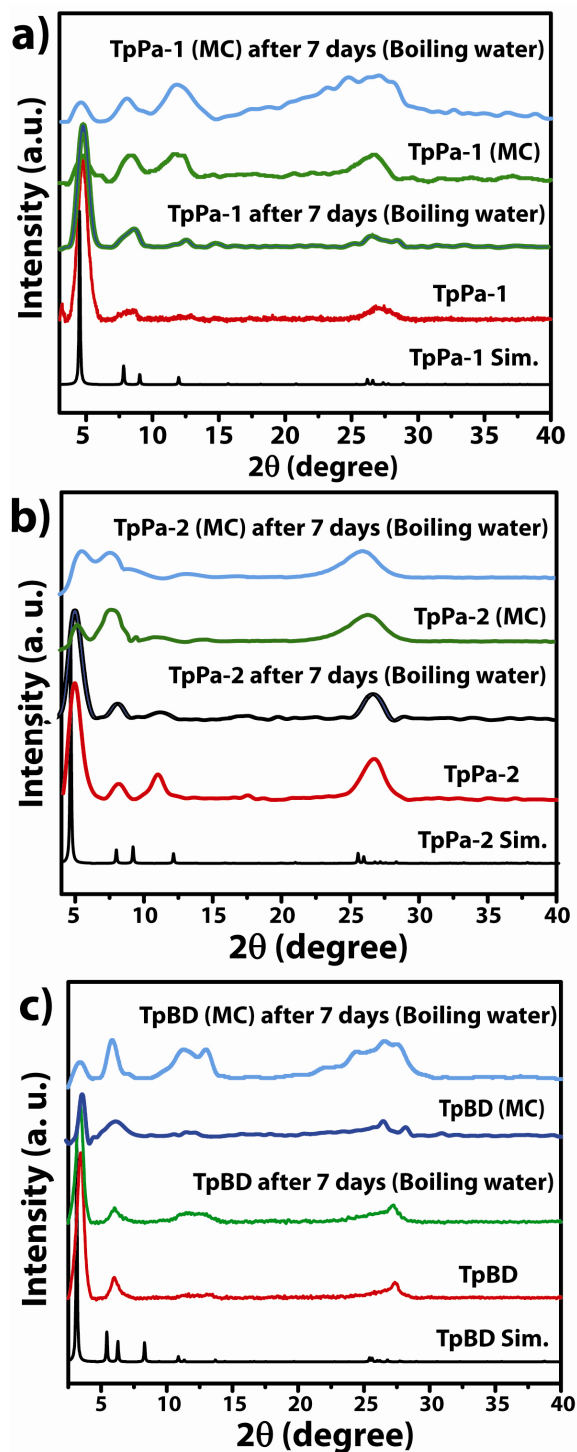


Figure S19: (a), (b) and (c) water stability test for COFs (MC) and COFs (ST) after dipping 7 days in boiling water.

Stability in Acid (HCl) and base (NaOH)

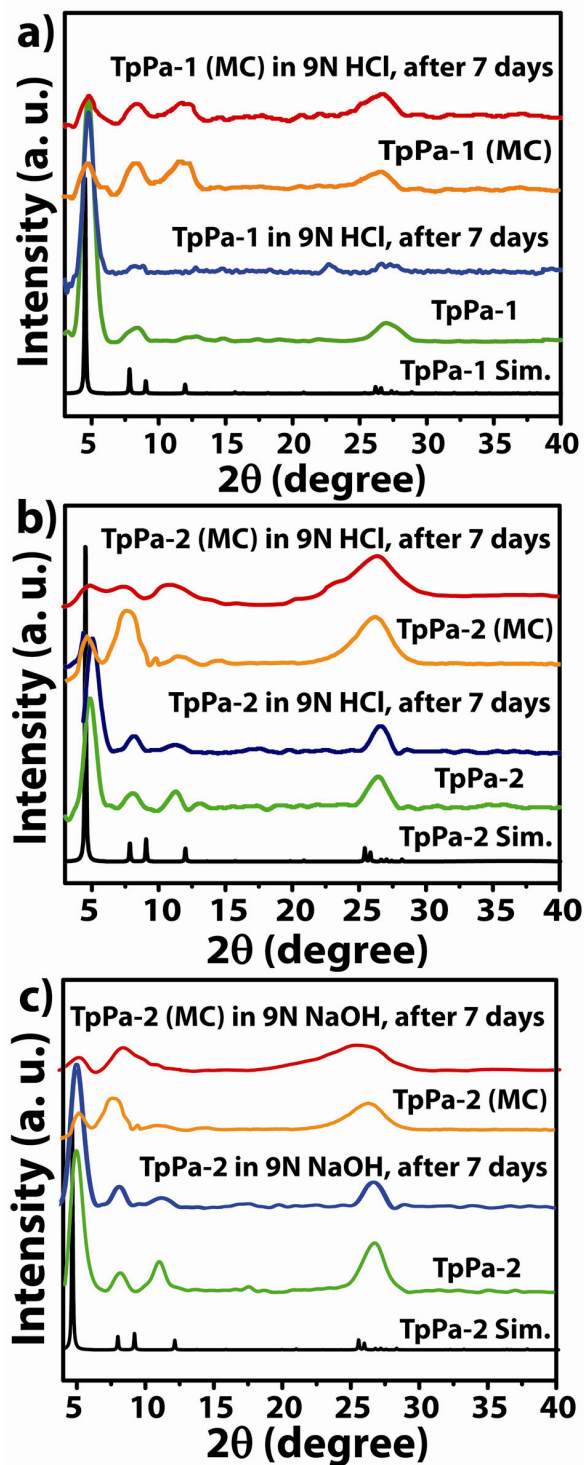


Figure S20: Acid stability test (a) **TpPa-1** (MC) (b) **TpPa-2** (MC) after dipping for 7 days in 9N HCl. (c) Base stability test of **TpPa-2** (MC) after dipping for 7 days in 9N NaOH.

Section S-11: Raman spectra

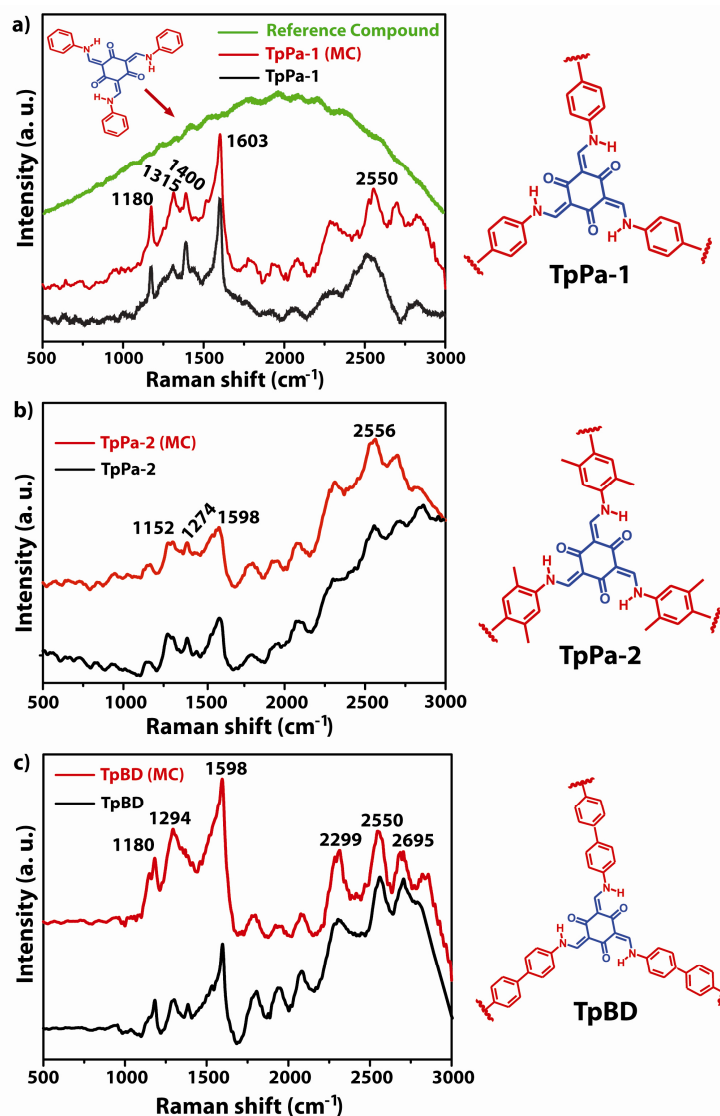


Figure S21: a, b and c; Raman spectra of all COFs (MC) and COFs (ST) measured at 514.5 nm.

We have collected the Raman spectra for the COF [(a) **TpPa-1** (black), **TpPa-1** (MC) (red) and reference compound (green); (b) **TpPa-2** (black) and **TpPa-2** (MC) (red); (c) **TpBD** (black) and **TpBD** (MC) (red)] samples synthesized via both mechanochemical (MC) as well as solvothermal methods to compare the structural similarity. It was found that there is no shifting of characteristic Raman peaks of the COFs. It is noteworthy that there were no Raman peaks for the reference compound (Monomer) (green, Figure S21 (a) at top) ($\lambda=514.5$ nm). So we believe that, the peaks appear in the case of COFs [**TpPa-1**, **TpPa-1** (MC), **TpPa-2**, **TpPa-2** (MC), **TpBD** and **TpBD** (MC)] are due to their overall structural attributes. However, we could not able to assign the peaks in the Raman spectra but the matching of these peaks indicates the retention of structural rigidity of COFs under synthesized by both methods.

Section S-12: PXRD of COFs (MC) synthesized using ball mill (MM400)

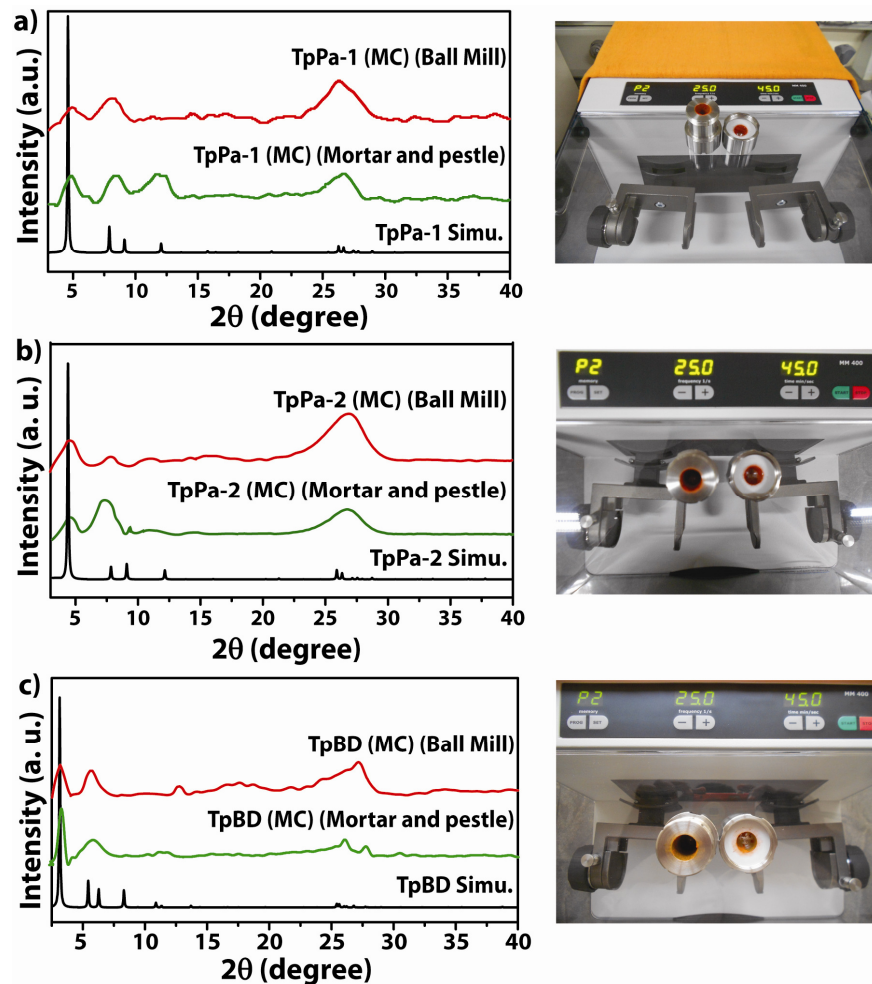


Figure S22: PXRD patterns (red) for the COFs (MC) [a) **TpPa-1** (MC) (Ball Mill) b) **TpPa-2** (MC) (Ball Mill) c) **TpBD** (MC) (Ball Mill)] synthesized using Retsch (MM400) Ball Mill operated at 25 Hz for 45 minutes; the green pattern corresponds to the COFs (MC) synthesized using mortar and pestle, finally both are compared with the simulated (black) one derived from the eclipsed stacking models of **TpPa-1**, **TpPa-2** and **TpBD**. Images on the right side of PXRD shows the colour of **TpPa-1** (MC), **TpPa-2** (MC) and **TpBD** (MC) respectively, synthesized using Ball Mill.

References:

- 1) Chong, J. H.; Sauer, M.; Patrick, B. O.; MacLachlan, M. J. *Org. Lett.* **2003**, 5, 3823.
- 2) Lukose, B.; Kuc, A.; Heine, T. *Chem. Eur. J.* **2011**, 17, 2388.
- 3) Accelrys, Material Studio Release Notes, Release 4.2, *Accelrys Software, San Diego* **2006**.